

Research engineer specializing in the co-design of machine learning algorithms and hardware accelerators. Expertise in device-circuit-ML integration and hands-on hardware prototyping.

### Education

- 08/2022–05/2027 **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00*  
(Expected) Advisor: Dr. Abhronil Sengupta
- 08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00* | Thesis: Neuromorphic Computing for Lifelong Learning
- 02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, *CGPA: 3.81/4.00*

### Academic Research and Teaching Experience

- 08/2022–Present **Graduate Research Assistant**, *Penn State*, University Park, PA
- Leading the design of a bit-serial ML accelerator for Transformer workloads, implementing CIM/PIM architectures with custom dataflow optimization for energy-efficient inference.
  - Pioneered device-circuit-ML co-design integrating spintronic NVM devices with hardware-aware training, demonstrating end-to-end flow from device fabrication to system-level deployment.
  - Developed brain-inspired ML systems including Astromorphic Transformer and RMAAT, with focus on hardware-aware model training and crossbar-aware quantization for CIM arrays.
- 08/2024–05/2025 **Graduate Teaching Assistant**, *Penn State*, State College, PA
- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
  - Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.
- 02/2021–08/2022 **Lecturer**, *University of Liberal Arts Bangladesh (ULAB)*, Dhaka, Bangladesh
- Taught undergraduate courses: Solid State Devices, Digital Circuit Design, Semiconductor Device Physics.

### Technical Skills

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|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ML Systems & Accel. | Hardware-aware ML; ML accelerator/ASIC design; neural network accelerator architecture; AI inference engine design; GPU architecture fundamentals; long-context sequence modeling; device-circuit-ML co-design; Transformer/LLM acceleration; model compression (PTQ/QAT, pruning, distillation); mixed-precision (FP16/INT8); dataflow optimization; PIM; near-memory computing |
| AI Frameworks       | PyTorch, TensorFlow, JAX, ONNX, TensorRT; model development and debugging; inference performance tuning; experiment tracking (W&B, Matplotlib); data tooling (Pandas, NumPy, Jupyter)                                                                                                                                                                                            |
| Hardware & EDA      | Cadence Virtuoso; Spectre; HSPICE; TCAD; COMSOL; ModelSim; Quartus; Vivado; Synopsys (Design Compiler, PrimeTime, VCS); ADC/DAC; mixed-signal design; CIM systems                                                                                                                                                                                                                |
| Digital/ASIC        | RTL design; CMOS/FinFET circuit design; SRAM design; sense amplifier design; memory peripheral circuits; FPGA prototyping; synthesis; P&R; functional verification; DFT; static timing analysis; timing closure                                                                                                                                                                  |
| Programming         | Python; CUDA; C/C++; MATLAB; Verilog; Shell/Bash; Linux/Unix; Git; Docker                                                                                                                                                                                                                                                                                                        |
| DTCO & STCO         | Design-technology and system-technology co-optimization; cross-layer PPA; process-aware design; system-architecture co-design                                                                                                                                                                                                                                                    |
| AI Dev.             | Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development                                                                                                                                                                                                                                                                               |
| Collaboration       | Cross-functional collaboration; mentoring; technical writing; Microsoft Office; Google Workspace                                                                                                                                                                                                                                                                                 |

## Industrial Experience

- 05/2025–07/2025 **Graduate Technical Intern, Intel Corporation, Hillsboro, OR**
- Developed AI-driven models for process-circuit co-optimization in ML accelerator hardware, focusing on yield prediction and performance enhancement.
  - Applied machine learning to optimize thin film deposition parameters affecting AI hardware circuit characteristics and system performance.
  - Utilized statistical analysis (JMP) and AI models to predict process impact on circuit electrical parameters and device reliability for ML hardware.
- 09/2020–07/2021 **R&D Engineer, SEMWAVES Ltd., London, UK, part-time**
- Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
  - Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

## Research Projects

- Astromorphic Transformer** Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture leveraging neuron-astrocyte interactions to emulate self-attention mechanisms. Incorporated Hebbian and presynaptic plasticities with non-linearities and feedback, achieving superior accuracy and faster convergence on sentiment classification (IMDB), image classification (CIFAR-10), and language modeling (WikiText-2) tasks. See publication: [IEEE TCDS 2025].
- Neuromorphic Cybersecurity with Lifelong Learning** Lead Student Researcher, 2023–2025. Developed a Hierarchical Dynamic Spiking Neural Network (D-SNN) for Network Intrusion Detection Systems combining static SNN detection (94.3% accuracy) with adaptive dynamic SNN classification using GWR-inspired structural plasticity and novel Adaptive STDP (Ad-STDP) learning. Achieved 85.3% overall accuracy on UNSW-NB15 benchmark in semi-supervised lifelong learning, demonstrating superior adaptation to new attack types while mitigating catastrophic forgetting (5.3% improvement over baseline). Demonstrated high operational sparsity suitable for neuromorphic hardware deployment. See publication: [arXiv], [ICONS 2025].
- MIPS Micro-processor Design** Lead Student Researcher, 2018–2019. Designed and implemented a 5-stage pipelined MIPS microprocessor in Verilog with instruction/data memory, forwarding, and hazard detection. Verified functionality through comprehensive simulation and synthesized for FPGA deployment.

## Select Publications

- [1] Md Zesun Ahmed Mia, Malyaban Bal, and Abhronil Sengupta. “Delving deeper into astromorphic transformers”. In: *IEEE Transactions on Cognitive and Developmental Systems* (2025).
- [2] Md Zesun Ahmed Mia et al. “Neuromorphic Cybersecurity with Semi-Supervised Lifelong Learning”. In: *Proceedings of the International Conference on Neuromorphic Systems (ICONS '25)*. IEEE Press, 2025, pp. 235–238. DOI: 10.1109/ICONS69015.2025.00043.
- [3] Md Zesun Ahmed Mia, Malyaban Bal, and Abhronil Sengupta. *RMAAT: A Bio-Inspired Approach for Efficient Long-Context Sequence Processing in Transformers*. 2024. URL: <https://openreview.net/forum?id=ikSrEv8FIId>.
- [4] Md Moinul Islam et al. “DCNN-LSTM based audio classification combining multiple feature engineering and data augmentation techniques”. In: *Intelligent Computing & Optimization: Proceedings of the 4th International Conference on Intelligent Computing and Optimization 2021 (ICO2021) 3*. Springer. 2022, pp. 227–236.

## Recognitions

- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024), Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

## Professional Affiliations

- Reviewer, Design Automation Conference (DAC) 2025, IEEE MWSCAS 2025, IACCESS 2024
- Student Member, IEEE (2015-Present), Executive Member, EDS, IEEE BD (2021-2022)